CUSTOMER SHOPPING BEHAVIOR

Strategic Analysis Report

Analysis of 3,900 Customer Transactions

Across Clothing, Footwear, Accessories & Outerwear

March 2026

1. Executive Summary

This report presents a comprehensive analysis of customer shopping behavior derived from a dataset of 3,900 purchase transactions. Using a multi-tool analytical pipeline — Python for data preparation, PostgreSQL for business-query execution, and Power BI for visualization — the study surfaces actionable insights around revenue drivers, product performance, customer loyalty, and subscription dynamics.

3,900	$59.76	3.75	27%
Total Transactions	Avg. Purchase Amount	Avg. Review Rating	Subscription Rate

Key findings at a glance:

- **Gender Revenue Gap:** Male customers generated $157,890 in total revenue — more than double the $75,191 from female customers — despite both groups comprising the same product catalogue.
- **Loyal Customer Dominance:** 80% of the customer base (3,116 of 3,900) qualify as Loyal (11+ previous purchases), signalling high retention but a narrow funnel for new acquisition.
- **Subscription Opportunity:** Subscribers represent only 27% of customers (1,053) yet exhibit near-identical average spend ($59.49) to non-subscribers ($59.87), suggesting untapped lifetime-value upside if subscription rates are improved.
- **Discount Dependency Risk:** Top products such as Hats (50%) and Sneakers (49.7%) show near-equal split between discounted and full-price purchases, creating margin pressure without a clear revenue premium.
- **Age Group Revenue:** The 20–29 (Young Adult) cohort drives the highest revenue at $62,143, followed closely by the 30–39 (Middle-aged) group at $59,197.

2. Methodology

The analysis followed a structured three-stage pipeline to ensure data integrity and reproducible findings.

2.1 Data Collection & Scope

The dataset comprises 3,900 transactional records with 18 attributes spanning customer demographics, item-level purchase details, and post-purchase behavioral signals. The time-agnostic nature of the dataset means all comparisons are cross-sectional rather than longitudinal.

2.2 Python — Data Preparation

- **Loading & Inspection:** Imported via pandas; used df.info() and .describe() to profile shape, types, and summary statistics.
- **Missing Data Treatment:** 37 null values identified in the Review Rating column were imputed using the median rating per product category, preserving category-level benchmarks without introducing global bias.
- **Column Standardisation:** All column names were converted to snake_case for downstream SQL compatibility.
- **Feature Engineering:** (a) age_group — continuous Age binned into four cohorts (Under 20, 20–29, 30–39, 40–49, 50+); (b) purchase_frequency_days — derived from the Frequency of Purchases text field to enable numeric comparison.
- **Redundancy Check:** promo_code_used was found to be a near-perfect proxy of discount_applied and was dropped to avoid multicollinearity in segmentation queries.
- **Database Export:** Cleaned DataFrame loaded into a PostgreSQL instance for SQL analysis.

```
In [32]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

In [33]:
```
df.describe()
```

Out[33]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

In [35]:
```
df.isnull().sum()
```

Out[35]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating             37
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [37]:
```
df.isnull().sum()
```

Out[37]:
```
Customer ID                0
Age                        0
Gender                     0
Item Purchased             0
Category                   0
Purchase Amount (USD)      0
Location                   0
Size                       0
Color                      0
Season                     0
Review Rating              0
Subscription Status        0
Shipping Type              0
Discount Applied           0
Promo Code Used            0
Previous Purchases         0
Payment Method             0
Frequency of Purchases     0
dtype: int64
```

In [41]:
```
df[['age', 'age_group']].head(10)
```

Out[41]:

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young Adult
8	26	Young Adult
9	57	Middle-aged

In [46]:

```python
import pandas as pd
from sqlalchemy import create_engine
```

In [48]:

```python
from sqlalchemy import create_engine
# Step 1: Connect to PostgreSQL
# Replace placeholders with your actual details
username = "postgres" # default user
password = "Sanika243$" # the password you set during installation
host = "localhost"
# if running locally
port = "5432"
# default PostgreSQL port
database = "customer_behavior" # the database you created in pgAdmin

engine = create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")

# Step 2: Load DataFrame into PostgreSQL
table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

print(f"Data successfully loaded into table '{table_name}' in database '{database}'.")
```

```
Data successfully loaded into table 'customer' in database 'customer_behavior'.
```

2.3 SQL — Business Query Layer

Ten structured queries were executed against the PostgreSQL database to answer discrete business questions. Techniques employed include aggregate functions (SUM, AVG, COUNT), correlated subqueries, CASE-WHEN classification, Common Table Expressions (CTEs), and window functions (ROW_NUMBER with PARTITION BY).

2.4 Power BI — Visualisation

An interactive dashboard was constructed in Power BI using the cleaned dataset. Slicers for Subscription Status, Gender, Category, and Shipping Type enable multi-dimensional filtering. KPI cards surface headline metrics; bar and donut charts display segment distributions and revenue breakdowns.

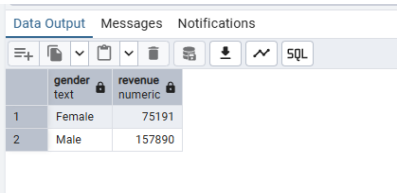
3. Key Findings

3.1 Revenue by Gender

Male customers contribute significantly more revenue in absolute terms, accounting for approximately 68% of total revenue. This is not necessarily a reflection of higher per-transaction spend — average purchase amounts are broadly similar — but rather that the dataset contains a larger proportion of male customers (approximately 2,652 of 3,900).

Gender	Total Revenue (USD)
Male	$157,890
Female	$75,191

> **Strategic Implication**
> The gender revenue gap is largely structural (volume-driven) rather than behavioural. Marketing investment in female customer acquisition and retention presents a direct path to revenue growth.



	gender text	revenue numeric
1	Female	75191
2	Male	157890

3.2 High-Value Customers Using Discounts

A subquery analysis identified 839 customers who simultaneously applied a discount and exceeded the dataset average purchase amount ($59.76). These customers represent a high-value cohort: they are deal-motivated but their basket size already exceeds the mean, indicating willingness and ability to spend above average even at full price.

> **Opportunity**
> Targeted A/B testing — offering this segment smaller discounts or loyalty points instead of blanket percentage-off promotions — could preserve revenue while sustaining engagement.

3.3 Top 5 Products by Review Rating

Rating differences across products are narrow (3.78–3.86), signalling broadly consistent customer satisfaction. Gloves, Sandals, and Boots lead the ranking.

Rank	Product	Avg. Review Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

The compressed rating range (0.08 spread across the top 5) suggests there are no strongly under-performing products but also no standout brand champions. Initiatives to cultivate five-star reviews for leading products could sharpen competitive positioning.

3.4 Shipping Type & Purchase Amount

Express shipping is associated with a modestly higher average transaction value ($60.48 vs. $58.46 for Standard). While the $2.02 difference appears small in isolation, it is consistent with a behavioural pattern where customers placing larger, more considered orders opt for faster delivery.

Shipping Type	Avg. Purchase Amount (USD)
Express	$60.48
Standard	$58.46

3.5 Subscriber vs. Non-Subscriber Spend

The subscription analysis reveals a surprising result: non-subscribers not only outnumber subscribers nearly 3:1 (2,847 vs. 1,053) but also generate 73% of total revenue ($170,436 vs. $62,645). Critically, average spend per transaction is nearly identical between the two groups.

Status	Customers	Avg. Spend (USD)	Total Revenue (USD)
Subscribed	1,053	$59.49	$62,645
Non-Subscribed	2,847	$59.87	$170,436

Key Insight

The subscription programme has not yet differentiated subscriber spend upward. This may indicate the current subscription offering lacks sufficient incentive. A redesigned programme with exclusive perks, early access, or tiered rewards could shift average subscriber spend above the non-subscriber baseline.

Data Output Messages Notifications

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

3.6 Products with High Discount Dependency

Five products show discount application rates approaching or exceeding 50% of all purchases, which poses a margin risk if the discounting is not deliberately strategic.

Product	Discount Rate (%)
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Data Output Messages Notifications

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

These products appear to be discount-dependent sellers — customers may have been conditioned to expect price reductions, which erodes perceived value and margins simultaneously.

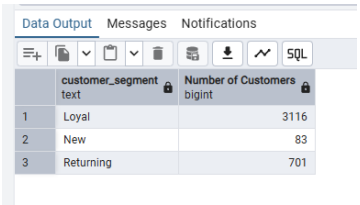
3.7 Customer Segmentation

Customers were classified into three segments based on previous purchase count: New (1 purchase), Returning (2–10 purchases), and Loyal (11+ purchases). The distribution reveals a highly mature base with limited new customer volume.

Segment	Customers	% of Base
Loyal	3,116	79.9%
Returning	701	18.0%
New	83	2.1%

Acquisition Alert

Only 2.1% of the current base are new customers. While high loyalty is a strength, the absence of new customer inflow creates long-term churn risk. Top-of-funnel investment is critical.



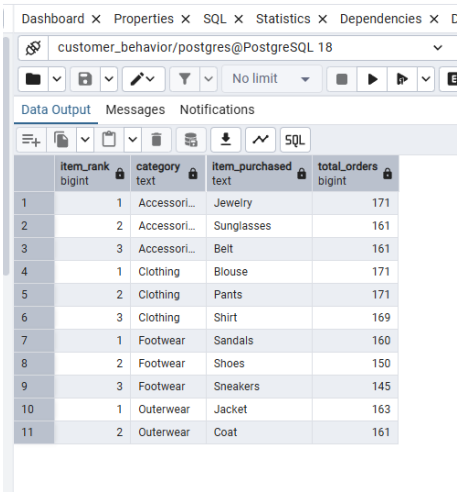
Data Output Messages Notifications

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

3.8 Top 3 Products per Category

Window function analysis identified the three highest-volume products within each of the four categories. Clothing and Accessories categories show the broadest competition among top-three products (Blouse, Pants, and Shirt are nearly tied; Jewelry, Sunglasses, and Belt are similarly matched).

Category	Rank	Product	Orders
Accessories	1	Jewelry	171
Accessories	2	Sunglasses	161
Accessories	3	Belt	161
Clothing	1	Blouse	171
Clothing	2	Pants	171
Clothing	3	Shirt	169
Footwear	1	Sandals	160
Footwear	2	Shoes	150
Footwear	3	Sneakers	145
Outerwear	1	Jacket	163
Outerwear	2	Coat	161
Outerwear	3	Gloves	155



	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

3.9 Repeat Buyers and Subscription Correlation

Among customers with more than 5 previous purchases (repeat buyers), non-subscribers still outnumber subscribers significantly (2,518 vs. 958). This suggests that high purchase frequency alone does not predict subscription uptake — the value proposition of subscribing must be more explicitly communicated to this group.

Subscription Status	Repeat Buyers (>5 purchases)
Non-Subscribed	2,518
Subscribed	958

3.10 Revenue Contribution by Age Group

Revenue is distributed relatively evenly across age groups, with Young Adults (20–29) leading marginally. The 50+ cohort (Senior) contributes nearly as much as the youngest group, dispelling assumptions about older customers being low-value segments.

Age Group	Total Revenue (USD)	% of Total
Young Adult (20–29)	$62,143	26.7%
Middle-aged (30–39)	$59,197	25.4%
Adult (40–49)	$55,978	24.0%
Senior (50+)	$55,763	23.9%

4. Power BI Dashboard

The findings above are presented through an interactive Power BI dashboard designed for real-time exploration. The dashboard supports slicing by Subscription Status, Gender, Product Category, and Shipping Type, enabling business stakeholders to drill into any combination of dimensions.

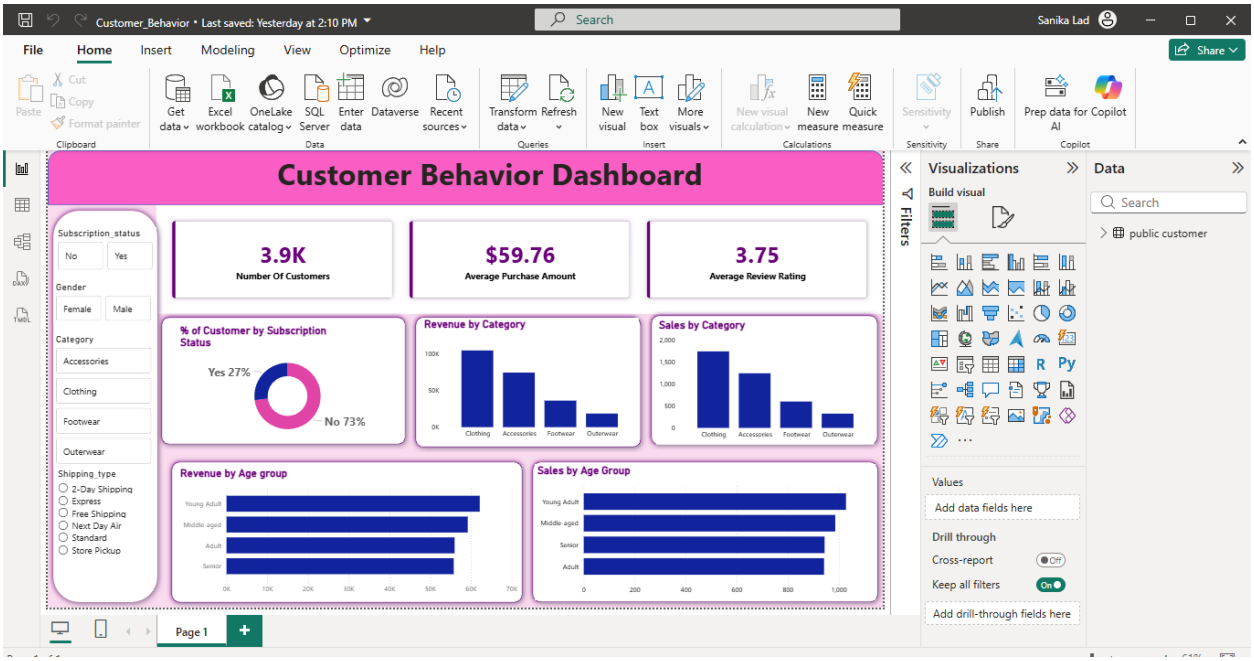


Figure: Customer Behavior Dashboard — Full view showing KPI cards, subscription distribution, revenue by category, and revenue/sales by age group.

Dashboard panels include:

- **KPI Cards:** Total customers (3.9K), Average Purchase Amount ($59.76), Average Review Rating (3.75).
- **Subscription Donut:** 27% Yes / 73% No — with slicer-linked drill-down by gender and category.
- **Revenue by Category:** Clothing leads, followed by Accessories, Footwear, and Outerwear.
- **Revenue & Sales by Age Group:** Horizontal bar charts comparing cohorts across both revenue and unit-sales dimensions.

5. Strategic Recommendations

5.1 Strengthen the Subscription Value Proposition

Current subscribers spend no more per transaction than non-subscribers. A redesigned loyalty programme should introduce differentiated benefits — such as free express shipping, early product access, or subscriber-only discounts — to drive incremental spend and justify enrolment.

Target metric: Increase avg. subscriber spend above $65 (vs. current $59.49) within two quarters of programme relaunch.

5.2 Invest in New Customer Acquisition

With only 83 new customers (2.1% of the base), the business faces a pipeline risk. Structured acquisition campaigns — particularly digital channels targeting the 20–29 demographic which already leads on revenue — should be prioritised. A referral programme leveraging the existing Loyal base could accelerate inflow at lower cost-per-acquisition.

5.3 Rationalise the Discount Framework

Hat, Sneakers, Coat, Sweater, and Pants are all approaching 50% discount application rates. A structured markdown calendar with pre-defined windows, rather than product-wide blanket discounting, would protect perceived value and improve gross margins. For high-value discount users (the 839 identified in Finding 3.2), a loyalty points alternative should be tested.

5.4 Activate Female Customer Revenue

Female customers represent a significant revenue growth opportunity — not through higher individual spend (already comparable to males) but through volume. Targeted campaigns, female-specific editorial content, and category-specific promotions (e.g., Accessories, which includes top-rated items) could close the revenue gap.

5.5 Leverage Product Quality Leaders in Marketing

Gloves, Sandals, and Boots lead in review ratings and therefore carry the strongest social proof. Feature these products prominently in acquisition campaigns, homepage merchandising, and email marketing to new or returning customers to maximise conversion rates.

5.6 Engage High-Frequency Non-Subscribers

2,518 repeat buyers (>5 purchases) remain unsubscribed. These customers are the most likely to convert given their demonstrated loyalty. A personalised outreach sequence — triggered after the 5th purchase — with an exclusive subscription offer could materially shift the 27% subscription rate upward.

6. Appendix — Analytical Methodology Summary

6.1 Data Pipeline Overview

Stage	Tool	Purpose
1. Ingestion	Python / pandas	Load CSV, profile structure, summary stats
2. Cleaning	Python / pandas	Null imputation (median by category), snake_case rename
3. Feature Eng.	Python / pandas	age_group bins, purchase_frequency_days
4. DB Load	Python / psycopg2	Push cleaned DataFrame to PostgreSQL
5. SQL Analysis	PostgreSQL	10 business queries (aggregations, CTEs, window funcs)
6. Visualisation	Power BI	Interactive dashboard with slicers and KPI cards

6.2 SQL Query Index

#	Query Objective	Technique Used
Q1	Revenue by Gender	GROUP BY + SUM
Q2	High-spending discount users	Correlated subquery + WHERE filter
Q3	Top 5 products by avg. rating	GROUP BY + AVG + ORDER BY + LIMIT
Q4	Standard vs. Express avg. spend	GROUP BY + AVG + WHERE IN
Q5	Subscriber vs. non-subscriber spend	GROUP BY + SUM + AVG + COUNT
Q6	Products by discount rate	CASE WHEN + SUM / COUNT ratio
Q7	Customer segmentation (CTE)	CTE + CASE WHEN + GROUP BY
Q8	Top 3 products per category	CTE + ROW_NUMBER OVER (PARTITION BY)
Q9	Repeat buyer subscription status	WHERE > 5 + GROUP BY
Q10	Revenue by age group	CASE WHEN age bins + SUM + GROUP BY

6.3 Dataset Attributes

18 columns across five domains: Customer Identity (Customer ID, Age, Gender, Location), Purchase Details (Item Purchased, Category, Purchase Amount USD, Season, Size, Color), Behavioural Signals (Review Rating, Discount Applied, Previous Purchases, Frequency of Purchases), Subscription & Payment (Subscription Status, Payment Method), and Logistics (Shipping Type).

— End of Report —

March 2026